

Binary Information

Binary Name => Simple crackme

Language => C/C++

Arch => x86x64

Platform => Unix/Linux

```
$ file crackme
```

```
crackme: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked,  
interpreter /lib64/ld-linux-x86-64.so.2,
```

```
BuildID[sha1]=1e0d463a818b87940b6b0f472c2ec6276af79583, for GNU/Linux 3.2.0,  
stripped
```

Basic Workflow

- The program takes user key input
- Compares user key input with its own original key
- If the comparison fails it prints **try again!**

Finding the key

```
$ ltrace ./crackme
```

```
printf("Enter key: ")
```

```
11
```

```
__isoc99_scanf(0x556905c42010, 0x7ffe0cb78620, 0, 0Enter key: sourav
```

```
)
```

```
= 1
```

```
strcmp("sourav", "bd4c217637bc828982c090b2de41b84d"...)
```

```
= 17
```

```
puts("try again!"try again!
```

```
)
```

```
= 11
```

```
+++ exited (status 0) +++
```

```
$ ./crackme
```

```
Enter key: bd4c217637bc828982c090b2de41b84d
```

```
good job!
```

I gave a random input at first and **ltrace** command was able to trace the library function calls made and we can see it loaded the user input as **args1** and the original key as **args2**. Of course we failed at first since we were just testing but on the next time I gave the original key value and was able to crack the crackme!!!

Decrypting the hash

The screenshot shows the 'Md5 Encrypt & Decrypt' website. On the left is a dark sidebar with navigation links: Home, Encrypt / Decrypt, Obfuscation, Ciphers, Conversion, Network, API, Contact, and FR / EN. The main content area has a title 'Md5 Encrypt & Decrypt' with a lock icon. Below the title is a promotional banner for '15% off on select luxury hotels with MakeMyTrip'. There are two input boxes: the left one contains the hash 'bd4c217637bc828982c090b2de41b84d' and the right one contains 'bd4c217637bc828982c090b2de41b84d : pass1785'. Below these boxes are 'Encrypt' and 'Decrypt' buttons. A large advertisement for the iPhone 17 is displayed, featuring the text 'iPhone 17 - Magic Starts Here!' and 'BUY NOW'. At the bottom, a section titled 'What is MD5 ?' explains the algorithm. A small 'Privacy - Terms' link is in the bottom right corner.

So the password is **pass1785**.

Objectives

- Find the key (DONE!!!) ✓
- Find the Easter egg and decrypt - <https://md5decrypt.net/en/> Have fun!) (DONE!!!) ✓